

# Tewodros Amberbir Habtegebrial

PHD CANDIDATE · COMPUTER VISION

RPTU Kaiserslautern-Landau, Gottlieb-Daimler-Straße 47, 67663 Kaiserslautern

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## Education

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### RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

#### PHD, COMPUTER SCIENCE

Jan. 2018 - Dec. 2023

- Advisor: Prof. Didier Stricker
- PhD Thesis: "Learning Renderable 3D Representations from Minimal Scene Specifications".

### RPTU Kaiserslautern-Landau

Kaiserslautern, Germany

#### MSC, COMPUTER SCIENCE

Nov. 2014 - Jan. 2017

- Advisors: Dr. Christian Bailer, Prof. Didier Stricker
- Thesis: "Novel-view Synthesis with Deep Convolutional Networks".  
I developed a novel image-based rendering technique using Siamese patch-matching networks.
- Major subject: Intelligent Systems
- Minor subject: Psycho-linguistics
- Overall Grade: 1.3

### Mekelle Institute of Technology

Mekelle, Ethiopia

#### BSC, ELECTRONICS AND COMMUNICATIONS ENGINEERING

Oct. 2007 - Aug. 2012

- Advisor: Dr. Workneh Gebeyehu
- Thesis: "Visual Object Tracking Using Particle Filtering".  
I worked on video object tracking using Monte Carlo methods such as Particle Filters.
- Overall Grade: 3.79/4.00

## Awards & Fellowships

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### Google PhD Fellowship

2021-2023

#### GOOGLE RESEARCH

Research area: Computer Vision

### CS Graduate Scholarship

2015-2021

#### RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Graduate Scholarship to pursue Master in Computer Science

### Merit Scholarship

2008-2012

#### MEKELLE INSTITUTE OF TECHNOLOGY

Mekelle, Ethiopia

Awarded to top 1% of highschool graduates

## Internships and Student Researcher Positions

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### Google Research

Remote

#### STUDENT RESEARCHER: WITH DR. STEFAN POPOV

Aug. 2022 - Mar. 2023

- Topic: "Multi-view Consistent Diffusion Models"

### Meta Reality Labs

Remote

#### INTERN: WITH DR. CHAKRAVARTY CHAITANYA, DR. FEDERICO PERAZZI

Jun. - Sep. 2021

- Topic: "Low-Light NeRF"

## Professional Experience

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### Research Group Lead

GERMAN RESEARCH CENTER FOR ARTIFICIAL INTELLIGENCE (DFKI)

Leading Research projects on Image-based Rendering, VR and 3D Reconstruction

Kaiserslautern, Germany

Nov. 2023 - Now

### Software Engineer

ICOG LABS

I worked on research projects in unsupervised learning from videos and single image collections

Addis-Ababa, Ethiopia

Oct. 2013 - Oct. 2014

### Assistant Lecturer

AKSUM UNIVERSITY

Network Analysis and Synthesis, Electromagnetic Field Theory

Aksum, Ethiopia

Aug. 2012 - Sep. 2013

## Teaching Experience

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### 3D Computer Vision

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Semester 2018-2022

### Graph Theory

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Summer Semester 2016

### Case-based Reasoning

RPTU KAISERSLAUTERN-LANDAU

Kaiserslautern, Germany

Winter Semester 2016

## Programming Languages

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Python, Lua, Matlab, C, C++

CUDA, PyCUDA

## Selected Publications

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*Text2MPI: Learning to Generate Multi-Plane Images from Text.* Tewodros **Habtegebrial**, Stefan Popov, Didier Stricker, Varun Jampani and Vittorio Ferrari. Under Review, 2023.

*SOMSI: Spherical Novel View Synthesis with Soft Occlusion Multi-sphere Images.* Tewodros **Habtegebrial**, Christiano Gava, Marcel Rogge, Stricker Didier and Varun Jampani. CVPR, 2022.

*Generative View Synthesis: From Single-view Semantics to Novel-view Images.* Tewodros **Habtegebrial**, Varun Jampani, Orazio Gallo, and Didier Stricker. NeurIPS, 2020.

*Fast View Synthesis with Deep Stereo Vision.* Tewodros **Habtegebrial**, Kiran Varanasi, Christian Bailer, and Didier Stricker. VISAPP, 2019.

*Deep Convolutional Networks For Snapshot Hyperspectral Demosaicking.* Tewodros **Habtegebrial**, Gerd Reis, and Didier Stricker. WHISPERS, 2019.

*Fast Feature Extraction with CNNs with Pooling Layers.* Christian Bailer, Tewodros **Habtegebrial**, Kiran Varanasi, and Didier Stricker. BMVC, 2018.

*SphereGlue: Learning Keypoint Matching on High Resolution Spherical Images.* Christiano Gava, Vishal Mukunda, Tewodros **Habtegebrial**, Federico Raue, Sebastian Palacio and Andreas Dengel. CVPR-Workshops, 2023

*Amharic Text Image Recognition: Database, Algorithm, and Analysis.* Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebyehu Belay and Didier Stricker. International Conference on Document Analysis and Recognition (ICDAR), 2019

*Factored Convolutional Neural Network For Amharic Character Image Recognition.* Birhanu Hailu Belay, Tewodros **Habtegebrial**, Marcus Liwicki, Gebyehu Belay and Didier Stricker. International Conference on Image Processing (ICIP), 2019